Reject Option to reduce False Prediction Rates for EEG-Motor Imagery based BCI

M Ganeshkumar, Kai Keng Ang and Rosa Q. So

***Abstract* – The Filter Bank Common Spatial Pattern (FBCSP) algorithm had been shown to be effective in performing multi-class Electroencephalogram (EEG) decoding of motor imagery using the one-versus-the-rest approach on the BCI Competition IV Dataset IIa. In this paper, we propose a method to reduce false detection rates of decoding through a rejection option based on the difference in the posterior probability computed by the Naïve Bayesian classifier. We applied the proposed approach on the BCI Competition IV Dataset IIa, and the results showed a decrease in the false detection rates from 34.6 % to 6.9%, while average decoded trials decreased from 93.2% to 34.2% using a rejection threshold between 0.1 and 0.9. We subsequently formulated a method to optimize the rejection threshold based on the maximum $F_{0.5}$ score. The optimal rejection threshold yielded an average decrease in false detection rate to 19.1% with an average of 67.5% of trials decoded. The results showed the feasibility of decreasing false detection rates at a cost of rejection. Nevertheless, the results suggest that the use of reject option (RO) may be used as a training feedback system to train subjects' overt and covert EEG control strategies for better (dexterity and safety) continuous control of external device.**

***Index terms* - Pattern recognition, Reject Option, BCI**

I. INTRODUCTION

One of the biggest caveats in EEG-based BCI is the difficulty to extract high signal resolution of motor imagery data from the subject [1]–[4]. Data collected from the surface of the scalp via EEG is mostly of low bandwidth signals from collective neural activity. The lower signal resolution collected via EEG compared to other invasive techniques creates an added difficulty for decoding algorithms to discern distinct motor imagery patterns. This contributes to a higher false detection rate (false positive and false negative predictions) causing the overall decoding accuracy to be slightly higher than chance level, even though subjects may be neurologically healthy [5], [6]. It is predicted that 15-30% of the population may be BCI Illiterates, individuals unable to modulate their ERD/ERS activity to produce a higher than chance decoding accuracy, even with extensive training [2]. This limits the mass population in having access to the range of EEG based BCI applications. However, this BCI systems offer advantages, such as ease of implementation and cost savings for the user.

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The EEG based BCI systems are good platforms to optimise the algorithm and to train individuals to use motor imagery to control simple applications. To expand the range of EEG-based BCI applications, overall decoding accuracy of EEG-based motor imagery needs to improve and can be achieved by reducing false prediction rates. A false prediction rate of more than 10% possess a high risk to the user, especially when controlling an external device such as a wheelchair.

Traditional EEG-based decoding algorithms have four main components 1) Data collection: Raw data loaded into the algorithm 2) Data segmentation: Break down the raw data into subset components using suitable filters such as CSP 3) Feature extraction: Identify unique class features in each trial 4) Classifier: Decision to categorise the data into the respective class [7], [8]. Reject Option (RO) as an additional post-processing component in the context of video and image identification, was discussed in several literature to improve the overall decoding accuracy [9]–[11]. These studies demonstrated that the decoding accuracy increased significantly after considering the results from the non-rejected decisions. Trials that were rejected were subject to manual annotation, which was deemed to be the cost of implementing a RO component.

Hence, the use of reject option seemed to be a viable implementation to increase overall accuracy by rejected false predictions for BCI applications. However, studies on the performance of a post processing component such as a reject option in this context of EEG-based BCI has yet to surface. Given the results from other studies, RO supplementation to EEG-based decoding might reduce false detection rates to ultimately increase decoding accuracy during continuous control while incurring the cost of rejecting several trials.

This paper aims to provide an approach to reduce false detection through the use of rejection thresholds to filter out decoded outputs that have a higher potential for misclassification [9]–[11]. This preliminary study will be extended to determine if reject option can be used as a training feedback system for subjects to develop their overt and covert control strategies to emulate distinct motor imagery patterns for optimal decoding by the BCI. The remainder of this paper is organised as follows. Section II briefly describes the RO incorporated into the FBCSP algorithm [7] and the dataset used [12]. Section III describes the results and the optimisation measure used. Section IV concludes with discussion of the result and a description of future study.

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II. REJECT OPTION ALGORITHM IN FBCSP DECODER AND DATASET USED

The Filter Bank Common Spatial Pattern (FBCSP) algorithm developed by Ang et al. scored the highest maximum kappa value of 0.57 for the BCI Competition IV Dataset IIa. Algorithms developed for the same datasets by Guangquan et al., Song et al., Coyle et al. and Wu et al., scored 0.52, 0.31, 0.30 and 0.29 Kappa values respectively. Hence, the FBCSP was one of the state-of-the-art algorithms.

The FBCSP algorithm utilised a Naive Bayesian Parzen window (NBPW) binary classifier to perform multi-class classification using the one-versus-rest (OVR) approach [7]. The binary classifier performed four iterations ($i = 4$) of OVR classification where in each iteration, each class assumed the class label (ω_o) while the other three assumed the class label (ω_R).

$$p(\omega_{O,i}|x) = \frac{p(x|\omega_{O,i})p(\omega_O)}{p(x)} \tag{1}$$

During each iteration, the classifier determined the posterior probability ($p(\omega_O|x)$ for each class using Bayes rule in (1) while the posterior probability for the rest of the class labels was $p(\omega_R|x) = 1 - p(\omega_O|x)$. The rejection threshold was implemented in each iteration by taking the difference in the posterior probabilities (2).

$$\lambda_i = |\, p(\omega_{O,i}|x) - p(\omega_{R,i}|x)\, | \tag{2}$$

The rejection threshold (Z) was used to compute the output rejection decision function $f(\omega_{O,i}|x)$. An output from each iteration was labelled to be rejected ($f(\omega_{O,i}|x) = 0$) if the degree of difference in posterior probability (λ_i) during each OVR iteration falls below the threshold Z. Predicted output from each iteration was labelled to be retained ($f(\omega_{O,i}|x) = 1$) if the difference in posterior probability λ_i was greater than Z (3).

$$f(\omega_{O,i}|x) = \begin{cases} 1, & \lambda_i \geq Z \\ 0, & \lambda_i < Z \end{cases} \tag{3}$$

After performing all four iterations, if a trial has one rejection label from the binary classification, the overall predicted class output after computing multi-class classification, is rejected. A predicted output from a trial is only retained if all four classification iterations do not carry the label for rejection. Several studies have proposed varying reject option algorithms [9]–[11], however, this algorithm was developed based on the Bayesian Decision Theory to specifically suit the context of the NBPW classifier [8]. The cost incurred is the reduction in the number of trials decoded, hence, not producing a predicted class label given an evaluation trial.

The false detection rates (FDR) were calculated by summing the false positive and false negative scores. The F_β optimisation measure was used to determine the optimal rejection threshold that balanced the decrease in correctly labelled predictions and FDR scores by using precision and recall scores (4). Subject specific optimal rejection threshold (Z) was determined based on the highest F_β score. Precision denotes the number of correctly classified positive labels (TP)

$$F_{\beta,Z}{}^{T} = \frac{(\beta^2 + 1)Precision^T \cdot Recall^T}{\beta^2 \cdot Precision^T + Recall^T} \tag{4}$$

divided by the number of labels the decoder labelled as positive (TP + FP). Recall denotes the number of correctly classified positive labels (TP) divided by the number of positive labels in the original data (TP + FN). $\beta = 0.5$ was used to increase the weightage of the gain in precision over the recall values. This was to emulate practical conditions where a device that operates with the correct instructions is preferred over 100% utility of the device, even if it was incorrect [13].

T denotes the type of measure used, μ = micro-averaging ($F_{0.5}{}^{\mu}$) and M = macro-averaging ($F_{0.5}{}^{\mu}$). $F_{0.5}{}^{M}$ score is an accurate measurement on the overall performance of the classifier. However, $F_{0.5}{}^{\mu}$ was also used as it was a more accurate measure when the class distribution in the prediction was not equal or when a class was under or over represented. Hence, both $F_{0.5}{}^{\mu}$ and $F_{0.5}{}^{M}$ performance optimisation measures were used (4) for the context of multi-class optimisation measures as suggested by literature [9], [15].

The BCI Competition IV dataset IIa [12] was used to test the reject option. The data comprised of nine BCI subjects (labelled 1-9) who performed four class motor imagery (left hand, right hand, foot and tongue). The data from each subject comprised of a training and evaluation set, both with 288 trials of EEG measurements from 22 electrodes. Each trial was labelled which was later used as the benchmark to assess the accuracy of prediction. Each subject's training data was used to train the FBCSP classifier before testing using the evaluation data using the synchronous setup [7] with a 0.1 degree of increment in the rejection threshold Z till 0.9.

III. RESULTS AND OPTIMISATION MEASURES

A. *Improvement in accuracy and decrease in the number of trials decoded.*

A rejection threshold of 0 indicates the absence of a reject option. The results in **Figure 1A** show the expected decrease in the percentage of total number of correctly predicted labels out of 288 trials for all subjects at a consistent rate. With an increasing threshold, RO removed greater number of correctly predicted labels based on the rejection decision function (3). Based on the nine subjects, the average percentage decrease in correctly labelled predictions range from 65.4% to 24.5% with rejection threshold (Z) of 0 to 0.9.

FDR is also observed to decrease consistently for all subjects (**Figure 1B**) at varying rates. The average decrease in FDR was from 34.6% to 5.25% using a rejection threshold (Z) of 0 to 0.9. Subjects 2, 5, 6 who had less than 55% correctly

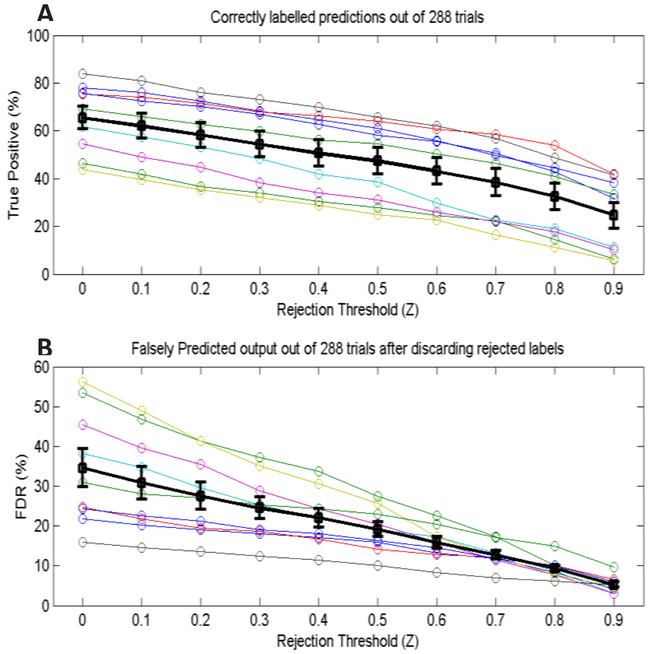

Fig. 1 Percentage change in correctly labelled predictions (A) and False Detection Rates (B) for each subject (Coloured) and on average (Black) with incremental increase in rejection threshold (Z = 0.1 to 0.9). Error bars indicate standard error across nine subjects.

labelled predictions showed FDR above 45% prior to RO. They showed less than 5% of FDR when Z = 0.9 while incurring cost of rejecting correctly labelled predictions. Subjects who initially had correct predictions over 70% showed a slower rate of decrease in FDR with RO. On average, the overall number of trials decoded out of 288 decreased from 100% to 34.2% as the rejection threshold increased from 0 to 0.9. Hence the optimisation measures were needed to balance the decrease in total predictions and FDR.

B. Optimisation measures to determine range of rejection thresholds for each subject

By taking nine subjects average, the optimal rejection threshold, which demonstrated the maximum $F_{0.5}^{\mu}$ and $F_{0.5}^{M}$ score was when Z = 0.5, shown in **Figure 2** with a red marker. Each individual subject displayed a range of optimal rejection threshold that indicated subject specific rejection threshold as showed in **Table 1**. Subjects 3, 4, 6, 8 and 9 demonstrated a maximum $F_{0.5}^{\mu}$ and $F_{0.5}^{M}$ score at the same rejection threshold while subjects 1, 2, 5 and 7 demonstrated a possible range of rejection thresholds. If the rejection threshold was set beyond the optimal values indicated in **Table 1**, the cost of rejecting predicted evaluation trials would overweigh the decrease in FDR, causing a decrease in $F_{0.5}$ measure. On average, with Z = 0.5, FDR was decreased to 19.1% while at the cost of decoding 67.5% of evaluation trial data.

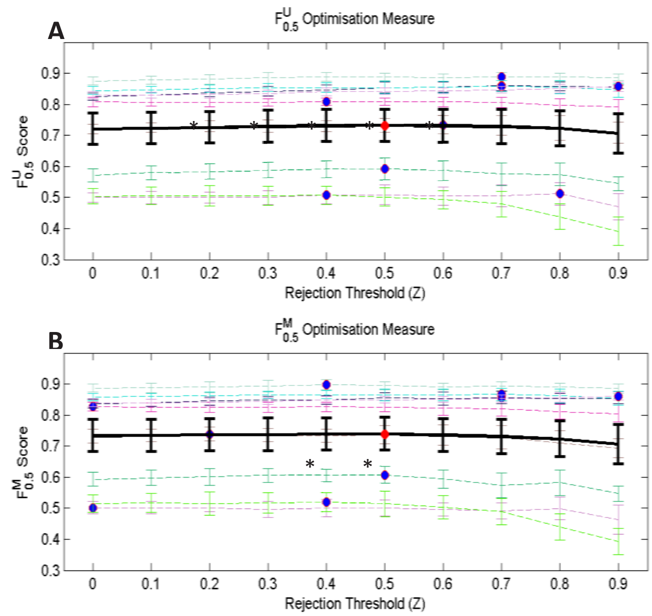

Fig. 2 Computation of (A) $F_{0.5}^{\mu}$ (micro-averaging) and (B) $F_{0.5}^{M}$ (macro-averaging) optimisation scores on average (Black) and individual nine subjects (Coloured) across the rejection threshold range 0.1 to 0.9. The red and blue markers indicate the highest $F_{0.5}^{\mu}$ and $F_{0.5}^{M}$ scores attained on average and for individual subjects respectively. Error bars indicate standard error across nine subjects. * compared against benchmark results, without RO (Z = 0), p<0.05, Wilcoxon Signed Rank Test.

IV. DISCUSSIONS AND FUTURE STUDY

Although the correctly labelled predictions (total True Positives out of 288 trials) decreased with an increase in rejection threshold, the False Detection Rate (FDR) decreased to less than 5% for all subjects at the highest rejection threshold. The overall cost of rejection incurred is the decrease in total number of predicted output which also includes the rejection of correctly labelled predictions. The optimisation measure is a valuable tool to determine the trade off point at which the cost of rejection supersedes the decrease in FDR. In the context of controlling an external device, a lower FDR is more favourable than a higher percentage of predicted output, catering to the safety of the user. In addition,

Table 1 Rejection thresholds (Z) at which each optimisation measure produced the highest score was obtained for each subject and on average. The corresponding percentage change in decoding accuracy and evaluation trials decoded at the respective optimal threshold (Z) is indicated

Subject	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**Avg**
Initial False Detection Rate (%)	21.9	53.5	24.7	38.2	45.5	56.3	16.0	24.3	30.9	**34.6**
Highest $F_{0.5}^{\mu}$ score attained (Z)	0.4	0.8	0.7	0.5	0.6	0.4	0.7	0.7	0.9	**0.5**
Highest $F_{0.5}^{M}$ score attained (Z)	0	0	0.7	0.5	0.2	0.4	0.4	0.7	0.9	**0.5**
New False Detection Rate (%)	19.4	16.0	12.0	17.5	17.5	23.8	12.0	11.8	6.9	**19.1**
Avg Decoded Eval Trials (%)	94.6	55.4	60.7	39.3	75.0	71.4	80.4	75.0	60.7	**67.5**

for the continuous control of an external device such as a wheelchair, the FBCSP employs a 2 second sliding window to decode the data to produce a predicted label output every 100ms [14]. According to the average results, there will be 67.5% of output when Z = 0.5. Hence, in a 1 second time frame, approximately 7 labels are produced to control a device while the FDR is reduced almost by half to 19.1% instead of the usual 10 predicted labels with a FDR of 34.6%. Given the reduction in the number of predicted label output, the dexterity of controlling the device might be compromised. On the other hand, this compromise could stand to be a boon for the user to build confidence and have a safer experience in controlling a device due to lower FDR.

Each subject displayed a variety of optimal rejection thresholds based on the optimisation measures which illustrates subject specific tuning parameters to maximise individual performance. This allows the conduct of personalised training session where each subject starts off at the optimal rejection threshold with higher safety (low FDR) but lower dexterity (low predicted output). Starting BCI training with the optimal rejection threshold might be useful to reduce a subject's anxiety and stress due to incorrect movement (low accuracy) and at the same time to reduce mental fatigue accumulated from the non-movement of the device despite performing motor imagery (high incidence of predicted trials rejected). Though studies have shown that moderate stress and fatigue levels improve EEG based BCI scores due to modifications in overt and covert controls [15]–[17], the decrease in incentive due to non-movement might supersede the motivation to optimise these control strategies.

As the subject improves in performing motor imagery patterns, the rejection threshold could be reduced to increase the number of predicted outputs and decrease the importance of reducing FDR. Such a training feedback system can motivate the user to fine tune overt and covert control strategies to achieve a higher amount of dexterity in controlling the device and safety aspects through lower FDR.

The use of reject option in EEG-based BCI also opens a new method to clean training data. RO can be used to identify poor data in training trials. These trials can be removed from the training data in order to train the classifier with a cleaner set of training data trials. Though, studies have shown that decoding accuracy decreased with a smaller quantum of training data [18], [19], there are methods to increase the training data given the same raw EEG dataset [20] or to improve the robustness of the decoder [21], which might be used as a supplement to increase decoding accuracy after removal of bad trial marks from the training data. In conclusion, the Reject Option shows a promising method to reduce False Detection Rates in EEG based Brain Computer Interface systems in order to control an external device with higher confidence.

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